

DD_MED-01 - Usage Mediation

Version: 1.0 Status: Draft for review Bundle: 05_billing Audience: billing developers, integration developers, usage rating developers, QA, NOC Related DDs:

- DD_PLM-CFG-07 - Voice Tariff Catalog
- DD_RAT-01 - Usage Rating
- DD_BIL-05 - Wallet and Topup
- DD_BIL-01 - Charging Engine Core
- DD_REP-01 - Reporting Data Mart and Dashboard API
- FE-APP-01 - BSS Backoffice Web App

MVP simplification baseline: MED-01 is offline/batch mediation only in this design. Implement after payment/wallet basics are stable. Do not introduce online charging mediation or real-time session control in the first release.

1. Purpose

MED-01 defines how SOPHIX receives raw usage records from external usage platforms, normalizes them into a common usage format, deduplicates them, quarantines invalid records, reconciles source totals, and emits trusted mediated usage records for rating.

The first required use case is voice CDR mediation from VoipSwitch or another SIP/voice platform.

MED-01 is the step before usage rating:

```
Voice platform CDR
-> MED-01 normalize/deduplicate/quarantine
-> RAT-01 rate and charge
-> BIL-05 wallet debit or BIL-01/BIL-02 postpaid charge
```

2. Ownership Boundary

Concern	Owner
Source CDR import from VoipSwitch/API/SFTP/manual file	MED-01
Raw CDR storage and audit	MED-01
CDR normalization to canonical fields	MED-01
Duplicate detection	MED-01
Quarantine and repair of invalid records	MED-01
Source-file/API reconciliation totals	MED-01
Tariff plans, zones, prefix tables, rates	PLM-CFG-07
Money calculation, allowance consumption, charge creation	RAT-01
Wallet debit for PREPAID rated usage	BIL-05 via RAT-01
POSTPAID usage charge/invoice integration	BIL-01/BIL-02 via RAT-01

Rule: MED-01 never calculates price and never debits money. A mediated record can be chargeable, zero-rated candidate, or rating-required, but monetary rating belongs to RAT-01.

3. Scope

3.1 In Scope

- Usage source registry.
- Raw CDR batch import.
- Raw CDR row storage.

- Canonical mediated usage record creation.
- CDR identity and idempotency hash.
- Duplicate handling.
- Basic technical validation:
 - required fields;
 - valid timestamps;
 - valid duration;
 - normalized calling/called numbers;
 - resolvable subscription/customer/service where possible.
- Quarantine queue for invalid/unresolved records.
- Manual repair and replay.
- Reconciliation of source batch counts and durations against imported/mediated/quarantined counts.
- Kafka events for mediated usage and quarantine.

3.2 Out Of Scope

- Tariff lookup and monetary calculation.
- Wallet debit.
- Invoice line creation.
- Tax calculation.
- Customer notification.
- Voice service provisioning.
- Vendor-specific switch configuration.

4. High Level Architecture

```

flowchart LR
    SRC["VoipSwitch / SIP platform"]
    IMP["MED Importer<br/>SFTP/API/manual"]
    RAW["Raw CDR Store"]
    NORM["Normalizer"]
    DEDUP["Dedup + validation"]
    MED["Mediated Usage"]
    QUAR["Quarantine"]
    RAT["RAT-01 Usage Rating"]
    BO["Backoffice Ops"]

    SRC --> IMP
    IMP --> RAW
    RAW --> NORM
    NORM --> DEDUP
    DEDUP --> MED
    MED --> RAT
    RAT --> BO
    BO --> NORM
  
```

5. Core Concepts

5.1 Usage Source

A usage source is an external platform that produces usage records.

Examples:

- VOIPSWITCH_KE
- SIP_PLATFORM_NRB
- future TV pay-per-view usage source
- future data usage platform

5.2 Raw CDR

A raw CDR is the original provider/source record stored with minimal transformation.

Raw CDRs must be retained for audit and reprocessing.

5.3 Mediated Usage Record

A mediated usage record is the canonical cleaned record consumed by RAT-01.

Example:

```
{
  "mediatedUsageId": "med-20260602-000001",
  "operatorCode": "WIK",
  "usageType": "VOICE_CALL",
  "subscriptionId": "SUB-700045",
  "serviceRef": "svc_VOIP_STD",
  "callingNumberNormalized": "+254700111222",
  "calledNumberNormalized": "+254711222333",
  "callStartedAt": "2026-06-02T10:15:00+03:00",
  "durationSeconds": 75,
  "sourceRecordId": "VS-CDR-998877",
  "mediationStatus": "READY_FOR_RATING"
}
```

5.4 Quarantine

Quarantine is the operational queue for records that cannot safely be mediated or rated.

Examples:

- missing subscription mapping;
- invalid timestamp;
- negative duration;
- missing called number;
- duplicate conflict;
- unsupported usage type.

6. Mediation Flow

6.1 Batch Import Flow

```
sequenceDiagram
    participant S as Usage Source
    participant I as MED Importer
    participant DB as MED DB
    participant M as Normalizer
    participant R as RAT-01

    S->>I: CDR file/API page
    I->>DB: Create raw usage batch
    I->>DB: Store raw CDR rows
    M->>DB: Load unprocessed raw CDR
    M->>M: Validate, normalize, deduplicate
    M->>DB: Insert mediated usage or quarantine
    M->>R: Emit UsageMediated event
```

6.2 Manual Repair Flow

```
flowchart TD
    A["Backoffice opens quarantine"]
    B["User corrects mapping or field"]
    C["MED validates repair"]
    D["Replay raw CDR"]
    E{"Valid now?"}
    F["Create mediated usage"]
    G["Keep quarantined with new note"]

    A --> B --> C --> D --> E
    E -->|Yes| F
    E -->|No| G
```

7. Business Rules

Rule ID	Rule	Notes
R-MED-01	Every imported CDR must be stored as raw data before normalization.	Prevents source data loss.
R-MED-02	Every raw CDR must have a deterministic dedupe hash.	Source id if stable, otherwise normalized field hash.
R-MED-03	Duplicate exact records are ignored after first successful mediation.	No double rating.
R-MED-04	Duplicate hash with conflicting payload is quarantined as DUPLICATE_CONFLICT.	Human review required.
R-MED-05	Missing required fields are quarantined, not dropped.	Data quality traceability.
R-MED-06	Records with unresolved subscription/customer are quarantined.	RAT-01 needs a billable target.
R-MED-07	Zero-duration calls can be mediated only if source disposition indicates answer or regulatory reason.	Avoid false chargeable records.
R-MED-08	Failed calls are not sent for rating unless product policy explicitly rates them.	Default no rating.
R-MED-09	Number normalization must be deterministic and operator-aware.	Kenya first, future regions configurable.
R-MED-10	Source batch reconciliation must compare source count/duration to stored/mediated/quarantined totals.	Operational control.
R-MED-11	Manual repair must preserve original raw payload and record user id.	Audit.
R-MED-12	Mediated records are immutable after emitted. Corrections create adjustment/reversal flow in RAT-01.	Avoid silent billing changes.

8. Data Model

8.1 usage_source

Stores external systems that provide usage records.

Use this table when onboarding VoipSwitch or another usage platform.

Column	Type	Purpose	Sample
usage_source_id	UUID PK	Internal source id.	usrc-001
operator_code	TEXT	Operator/tenant.	WIK
source_code	TEXT	Stable source code.	VOIPSWITCH_KE
source_type	TEXT	VOIPSWITCH, SIP_SWITCH, DATA_PLATFORM, TV_PLATFORM.	VOIPSWITCH
ingestion_mode	TEXT	SFTP_FILE, API_PULL, API_PUSH, MANUAL_UPLOAD.	SFTP_FILE
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE
timezone	TEXT	Source timestamp timezone if not explicit.	Africa/Nairobi
config_json	JSONB	Non-secret source config.	{"filePattern":"cdr_*.csv"}
secret_ref	TEXT NULL	Secret manager reference.	secret/med/voipswitch-ke
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T10:00:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T10:00:00Z

Recommended indexes:

- Unique (operator_code, source_code).
- (operator_code, status).

8.2 raw_usage_batch

Stores one imported file, API page group, or manual upload.

Use this table to reconcile source totals and track import status.

Column	Type	Purpose	Sample
batch_id	UUID PK	Batch id.	rub-001
usage_source_id	UUID FK	Source system.	usrc-001
operator_code	TEXT	Operator/tenant.	WIK
source_batch_ref	TEXT	File name, API page window, or upload id.	cdr_20260602_1000.csv
period_start	TIMESTAMP	Source period start.	2026-06-02T10:00:00+03:00
period_end	TIMESTAMP	Source period end.	2026-06-02T10:59:59+03:00
status	TEXT	RECEIVED, IMPORTING, IMPORTED, PARTIAL, FAILED, RECONCILED.	IMPORTED
source_record_count	INTEGER NULL	Count declared by source.	10000
source_total_duration_seconds	BIGINT NULL	Duration declared by source.	450000
stored_record_count	INTEGER	Raw rows stored.	10000
mediated_record_count	INTEGER	Rows mediated.	9950
quarantine_record_count	INTEGER	Rows quarantined.	50
file_checksum	TEXT NULL	File checksum for duplicate import detection.	sha256:abc...
created_at	TIMESTAMP	Import timestamp.	2026-06-02T11:05:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-06-02T11:10:00Z

Recommended indexes:

- Unique (usage_source_id, source_batch_ref).
- (operator_code, status, period_start).

8.3 raw_usage_record

Stores one raw CDR row.

Use this table for audit, replay, and exact duplicate detection.

Column	Type	Purpose	Sample
raw_record_id	UUID PK	Internal raw record id.	rur-001
batch_id	UUID FK	Import batch.	rub-001
usage_source_id	UUID FK	Source system.	usrc-001
operator_code	TEXT	Operator/tenant.	WIK
source_record_id	TEXT NULL	Source CDR id if present.	VS-CDR-998877
source_sequence_no	BIGINT NULL	Source/file row number.	42
record_hash	TEXT	Deterministic hash of identity fields.	sha256:def...
raw_payload_json	JSONB	Original row as parsed.	{"caller": "0700111222"}
processing_status	TEXT	NEW, MEDIATED, DUPLICATE, QUARANTINED, FAILED.	MEDIATED
created_at	TIMESTAMP	Stored timestamp.	2026-06-02T11:05:01Z
processed_at	TIMESTAMP NULL	Processing timestamp.	2026-06-02T11:05:05Z

Recommended indexes:

- Unique (usage_source_id, record_hash).
- (batch_id, processing_status).
- (operator_code, processing_status, created_at).

8.4 mediated_usage_record

Stores one normalized usage record ready for rating.

Use this table as the source for RAT-01.

Column	Type	Purpose	Sample
mediated_usage_id	UUID PK	Mediated record id.	med-001
raw_record_id	UUID FK	Raw source record.	rur-001
operator_code	TEXT	Operator/tenant.	WIK
usage_type	TEXT	VOICE_CALL, future DATA_USAGE, TV_EVENT.	VOICE_CALL
subscription_id	TEXT	Resolved subscription.	SUB-700045
customer_id	TEXT	Resolved customer.	CUS-100045
account_id	TEXT NULL	Billing account if available.	ACC-900045
service_ref	TEXT	PLM service.	svc_VOIP_STD
package_ref	TEXT NULL	Package active at call time.	pkg_INET_VOICE_STD
calling_number_normalized	TEXT NULL	Caller number in canonical form.	+254700111222
called_number_normalized	TEXT NULL	Called number in canonical form.	+254711222333
call_direction	TEXT	OUTBOUND, INBOUND, FORWARDED.	OUTBOUND
call_started_at	TIMESTAMP	Call start time.	2026-06-02T10:15:00+03:00
call_ended_at	TIMESTAMP NULL	Call end time.	2026-06-02T10:16:15+03:00
duration_seconds	INTEGER	Actual duration from source.	75
disposition	TEXT	Source-normalized call result.	ANSWERED
mediation_status	TEXT	READY_FOR_RATING, SENT_TO_RATING, RATED, RATING_FAILED.	READY_FOR_RATING
rated_usage_id	TEXT NULL	RAT-01 output id when rated.	rated-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-02T11:05:05Z

Recommended indexes:

- Unique (raw_record_id).
- (operator_code, mediation_status, created_at).
- (subscription_id, call_started_at).

8.5 mediation_quarantine

Stores raw records that cannot be safely mediated.

Use this table for operations repair.

Column	Type	Purpose	Sample
quarantine_id	UUID PK	Quarantine id.	mq-001
raw_record_id	UUID FK	Raw record.	rur-002
operator_code	TEXT	Operator/tenant.	WIK
reason_code	TEXT	Why quarantined.	SUBSCRIPTION_NOT_FOUND
severity	TEXT	LOW, MEDIUM, HIGH.	HIGH
status	TEXT	OPEN, IN_REVIEW, RESOLVED, IGNORED.	OPEN
repair_payload_json	JSONB NULL	User-provided correction.	{"subscriptionId":"SUB-700045"}
assigned_to_user_id	TEXT NULL	Backoffice assignee.	usr-billing-001
resolution_notes	TEXT NULL	Repair/ignore notes.	Mapped old DID to subscription
created_at	TIMESTAMP	Creation timestamp.	2026-06-02T11:05:10Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	NULL

Recommended indexes:

- (status, reason_code, created_at).
- (operator_code, severity, status).

8.6 mediation_reconciliation_run

Stores reconciliation for a source batch or time window.

Use this table to verify that imported/mediated/quarantined records match source totals.

Column	Type	Purpose	Sample
run_id	UUID PK	Run id.	mrr-001
operator_code	TEXT	Operator/tenant.	WIK
usage_source_id	UUID FK	Source reconciled.	usrc-001
batch_id	UUID NULL	Batch if batch-level.	rub-001
period_start	TIMESTAMP	Reconciliation start.	2026-06-02T10:00:00+03:00
period_end	TIMESTAMP	Reconciliation end.	2026-06-02T10:59:59+03:00
status	TEXT	RUNNING, COMPLETED, FAILED, PARTIAL.	COMPLETED
source_count	INTEGER	Count from source.	10000
stored_count	INTEGER	Raw rows stored.	10000
mediated_count	INTEGER	Mediated rows.	9950
quarantine_count	INTEGER	Quarantined rows.	50
mismatch_count	INTEGER	Count of discrepancies.	0
started_at	TIMESTAMP	Run start.	2026-06-02T11:20:00Z
completed_at	TIMESTAMP NULL	Run completion.	2026-06-02T11:21:00Z

Recommended indexes:

- (operator_code, status, period_start).
- (usage_source_id, period_start).

9. APIs

9.1 Source And Batch APIs

```
GET /api/usage-mediation/sources
POST /api/usage-mediation/sources
GET /api/usage-mediation/batches
GET /api/usage-mediation/batches/{batch_id}
POST /api/usage-mediation/batches/manual-upload
POST /api/usage-mediation/batches/{batch_id}/reprocess
```

9.2 Mediated Usage APIs

```
GET /api/usage-mediation/records?status=READY_FOR_RATING
GET /api/usage-mediation/records/{mediated_usage_id}
POST /api/usage-mediation/records/{mediated_usage_id}/mark-sent-to-rating
POST /api/usage-mediation/records/{mediated_usage_id}/mark-rated
```

9.3 Quarantine APIs

```
GET /api/usage-mediation/quarantine?status=OPEN
GET /api/usage-mediation/quarantine/{quarantine_id}
POST /api/usage-mediation/quarantine/{quarantine_id}/repair-and-replay
POST /api/usage-mediation/quarantine/{quarantine_id}/ignore
```

9.4 Reconciliation APIs

```
POST /api/usage-mediation/reconciliation-runs
GET /api/usage-mediation/reconciliation-runs
GET /api/usage-mediation/reconciliation-runs/{run_id}
```

10. Events

Event	Topic	When emitted
RawUsageBatchImported	sophix.usage.raw-batch.imported	Batch import completes.
UsageMediated	sophix.usage.mediated	One canonical usage record is ready for rating.
UsageQuarantined	sophix.usage.quarantined	Raw CDR cannot be mediated.
UsageMediationReplayed	sophix.usage.mediation.replayed	Quarantined record repaired and replayed.
UsageMediationReconciliationCompleted	sophix.usage.mediation.reconciliation.completed	Source batch/window reconciled.

Sample UsageMediated:

```
{
  "eventType": "UsageMediated",
  "operatorCode": "WIK",
  "mediatedUsageId": "med-20260602-000001",
  "usageType": "VOICE_CALL",
  "subscriptionId": "SUB-700045",
  "customerId": "CUS-100045",
  "serviceRef": "svc_VOIP_STD",
  "packageRef": "pkg_INET_VOICE_STD",
  "calledNumberNormalized": "+254711222333",
  "callDirection": "OUTBOUND",
  "callStartedAt": "2026-06-02T10:15:00+03:00",
  "durationSeconds": 75
}
```

11. Frontend Usage

FE-APP-01 Backoffice uses MED-01 for:

- Usage source configuration.
- Batch import dashboard.
- Raw batch detail and import counts.
- Quarantine queue.
- Repair and replay actions.
- Reconciliation run status.
- Deep link from customer/subscription to mediated usage history.

Self-care should not read MED-01 directly. Customer-visible usage history should come from RAT - 01 or a reporting/read API after records are rated and safe to display.

12. Infra Recommendation

12.1 Medium Budget Deployment

Component	Recommendation
API pods	2 pods, 1 vCPU, 1.5 GB RAM each.
Import worker	1 worker, 1 vCPU, 2 GB RAM.
Mediation worker	1-2 workers, 2 vCPU, 4 GB RAM each for CDR bursts.
DB	Billing PostgreSQL schema usage_med; partition raw and mediated records by month.
Object storage	Store original files in local object storage/S3-compatible bucket.
Queue/Kafka	Use Kafka or queue for batch import and mediation jobs.

12.2 Recommended Partitions

Partition monthly:

- raw_usage_record
- mediated_usage_record
- mediation_quarantine

Batch and source tables do not need partitioning at launch.

12.3 OS Recommendation

Use Ubuntu Server 24.04 LTS for VPS deployments or hardened Debian/Ubuntu slim base images for containers.

Workers should run close to the database and object storage. SFTP/API pull jobs should run from private network where possible.

13. AWS Deployment Recommendation

AWS component	Usage
ECS/EKS	Run mediation API and workers.
RDS PostgreSQL	Raw, mediated, quarantine, reconciliation tables.
S3 with KMS	Store original CDR files.
MSK or managed Kafka equivalent	Usage mediated/quarantine events.
Secrets Manager	SFTP/API credentials for source platforms.
CloudWatch	Import failures, worker lag, quarantine rate alarms.

Required alarms:

- Source batch not received within expected window.
- Quarantine rate above threshold.
- Mediation worker backlog growing.
- Duplicate conflict spike.
- Reconciliation mismatch count above threshold.

14. Testing Requirements

14.1 Unit Tests

- Parse VoipSwitch CDR row.
- Normalize local Kenya MSISDN to E.164.
- Build deterministic record hash.
- Reject missing called number.
- Reject negative duration.
- Quarantine unresolved subscription.
- Exact duplicate does not create second mediated record.
- Duplicate conflict creates quarantine.

14.2 Integration Tests

- Import file creates batch and raw records.
- Mediation worker creates mediated records.
- UsageMediated event is emitted.
- Quarantine repair and replay creates mediated record.
- Batch reconciliation totals match imported rows.

14.3 End To End Tests With RAT-01

- Raw answered voice CDR becomes mediated usage and is rated by RAT-01.
- Failed call is not sent to rating by default.
- Unknown subscription stays quarantined until repaired.
- Duplicate CDR is not rated twice.

15. Definition Of Done

MED-01 is complete when:

- Usage source configuration exists.
- Raw CDR batches can be imported from at least one source mode.
- Raw CDR rows are stored before processing.
- Normalization creates canonical mediated usage records.
- Dedupe prevents double mediation.
- Invalid records go to quarantine.
- Repair and replay works.
- UsageMediated events are consumed by RAT-01.
- Reconciliation compares source, stored, mediated, and quarantined totals.
- Backoffice can inspect batches and quarantine.